26DOC52F

FLOAT SPEC SHEET

FOR GYRODOS ROBOTICS' FLOAT MAGIKARP

La Salle College Robotics Team

Gyrodos ROBOTICS

EST. 2024 • HKG

26MEC34

FLOAT MAGIKARP

BY THE NUMBERS

Mass: 6kg (59 Newtons)
Dimensions: 1000(L)*160(W)*160(H) mm
Development Time: ~300 man-hours
Cost: USD $1275 (HK$ 9981)
Team size: 24 (6 returning)
2nd Model of Gyrodos Robotics' 2026 lineup
Power supply: 8*1.5V NiMH batteries in series

OPERATING SAFETY FEATURES

- 3A Fuse on Circuit
- Double-sealed Interior
- No Sharp Edges

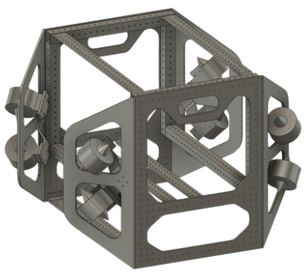
ROV DESIGN FEATURES

SPECIAL FEATURES

- [User-ex] Data Packet Automatically Broadcasts after dive is finished
- [Adaptability] PID Controlled Descent and Ascent
- [Efficiency] 500cm^3 syringe for greater max. acceleration

HOW IT WORKS

The buoyancy engine consists of a 500cc syringe connected to a vertical actuator which has 15cm of movement range. A H-bridge allows the onboard Arduino Nano to control the motor's power output. The Arduino Nano reads the current depth from the depth sensor, evaluates its position, and uses a PID (Proportional-Integral-Derivative) control algorithm to ensure the Float can make use of its maximum 0.36ms^{-2} acceleration to reach the target depth.



SPARKS SOLUTIONS
中山市三禾电子科技有限公司

MAKERHOLIC